



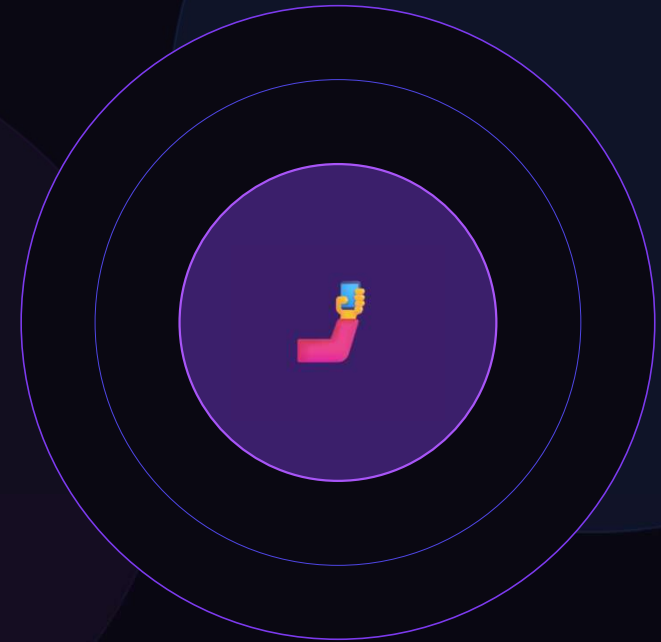
Face Find

Find Your Photos. Instantly.

AI-powered face recognition for event photography.

Guests scan a QR code, take a selfie — and find every photo of themselves in seconds.

• Team - 7



TEAM

Member Roles & Responsibilities.

Jash Berawala

Full Stack Lead

- Backend API (Node.js/Express)
- GitHub Management
- ML Integration
- API Development & Testing

Jay Shah

Frontend Developer

- React.js Application
- Admin Dashboard Interface
- Guest Photo Gallery
- Responsive Design & State Mgmt

Niraj Patil

ML Engineer

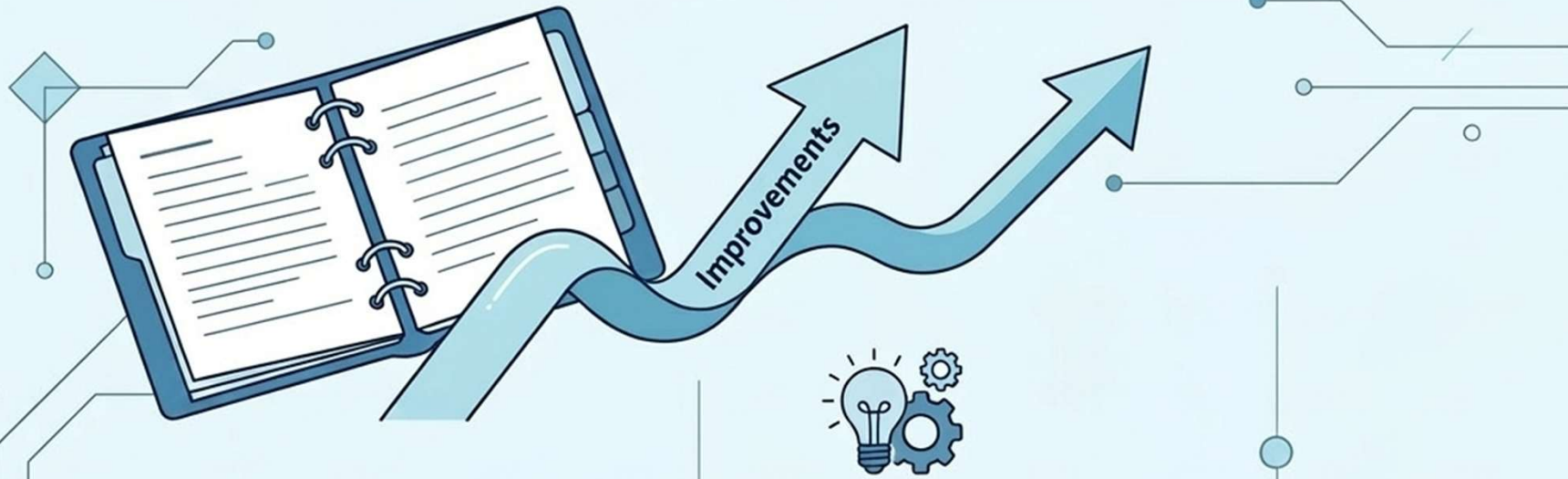
- Face Detection & Recognition
- Python Microservice (FastAPI)
- Face Clustering (DBSCAN)
- Model Performance Optimization

Chetan Patel

Database Administrator

- AWS S3 Integration
- Database Architecture (MongoDB)
- Docker Containerization
- Security & QR Code System

Improvements made from Professor's Feedback



- Project Description made more descriptive
- Changes made in Personas
- Error correction in Technologies(described)

THE PROBLEM

Event photos are impossible to find.



Thousands of photos

Photographers upload 1,000–5,000+ images per event.



No way to search

Guests have no tool to find photos of themselves.



Hours wasted

Manual searching kills the joy of the moment.

“

I spent 2 hours scrolling through 3,000 wedding photos just to find 12 pictures of myself.

— Event Attendee

94%

of guests never find all their photos

THE SOLUTION

FaceFind.

Scan. Snap. See yourself.

0 1



Photographer Uploads

Creates an event, uploads all photos.
AI processes and indexes every face
automatically.

0 2



Guest Scans QR

Attendees scan the unique QR code at
the venue to access the guest portal
instantly.

0 3



Takes a Selfie

Upload or take a selfie directly in the
browser — no app download
required.

0 4



Instant Match

AI matches your face against all event
photos and returns your results in
under 5 seconds.

PRODUCT

Built for Photographers, Loved by Guests.



AI Face Indexing

Every uploaded photo is automatically processed and indexed by face embedding.



< 5 Second Match

Guests receive matched results in under 5 seconds using vector similarity search.



No App Required

Fully browser-based. Guests just scan a QR code — nothing to install.



Photographer Auth

Secure JWT-based login. Each photographer sees only their own events.



Live Dashboard

Real-time processing updates via WebSockets. See photos processed as they upload.



QR Code Sharing

Auto-generated QR codes per event. Share with guests in seconds.

Our Commitment to Each Other.

1. Communication

- Primary platform: Microsoft Teams / Google Meet
- Respond to messages within 12 hours
- Blockers & delays communicated promptly and transparently

2. Meetings

- Attend all sprint ceremonies: Planning, Backlog Grooming ,Reviews
- Arrive prepared with updates and blockers
- Retrospectives conducted with candor and respect

3. Work Expectations

- Complete assigned tasks before the deadline
- Avoid last-minute submissions
- Submit work for peer review before merging to main
- Communicate blockers early so the team can adjust

Personas





Demographics

Name : Alex Reynolds

Age: 31

Location: Denver, CO

Business: Moments by Alex Photo

Income: \$65-85K/year

Profile

Full-time Event Photographer

8 years experience

Shoots 3-4 events/week

800-1200 photos per event

Specializes in weddings & corporate events

Goals

Reduce photo delivery time from hours to minutes

Improve client satisfaction with faster turnaround

Focus more on photography and client acquisition

Increase bookings through referrals

Build brand via photo tags

Quote

"I spent 6 hours last night just organizing and sending photos from Saturday's wedding. That's time I could have been shooting, marketing, or actually living my life. If guests could just scan a QR code and instantly get their photos, I'd be a hero."

— Alex Reynolds, Professional Photographer

Frustration

Manual folder creation and link sharing takes 5-6 hours per event

Constant client requests: "When will photos be ready?"

No easy way to provide partial access during events

Link sharing management is cumbersome

Lost referral opportunities from untagged photos

How FaceFind Helps

Creates event in 30 seconds

AI handles photo distribution

Guests self-serve via QR code

Real-time processing during breaks

Automatic tagging for referrals

Frees 5+ hours per event



Demographics

Name : **Marcus Thompson**

Age : 28

Location : Atlanta, GA

Occupation : Marketing Manager

Education : MBA, Emory University

Income : \$82,000/year

Profile

Marketing Manager with 5+ years experience

Attends 8-10 weddings annually

Active on Instagram & LinkedIn

Posts within 24 hours for max engagement

Values professional quality photos

Goals

Get photos immediately while memories are fresh

Easily find photos of himself without scrolling

Share to social media during the event

Enjoy events without phone distraction

Build social media presence

Quote

"I just got the photo link from my cousin's wedding THREE WEEKS after the event. The moment has passed, everyone's moved on. Scrolling through 900 photos to find the 3 where I look decent is exhausting. Why can't someone just send me MY photos?"

— Marcus Thompson, Wedding Guest

Frustration

Receives photo links 3-4 weeks after events. Must scroll through 800+ photos to find himself Misses social media engagement window. Phone photos are poor quality. Takes photos during events instead of enjoying

How FaceFind Helps

Scan QR code at event entrance

Upload one selfie

AI finds ALL photos of Marcus

Instant access to professional photos

Share during event when engagement peaks

Selfie deleted immediately after matching



Profile

Senior Events Manager with 12 years experience Organizes 15-20 corporate events annually Manages \$3M annual event budget Average event: 100-500 attendees Reports to C-suite leadership Data-driven decision maker

Goals

Quick, secure photo delivery to employees Track engagement and measure ROI Control access with email whitelisting Generate reports for leadership Ensure data privacy and compliance Demonstrate event value beyond attendance

Quote

"Last year's leadership summit cost \$250,000. When the CFO asked what we got for it, all I had was attendance numbers and some nice photos. Now I can show exactly how many people engaged, shared, and downloaded. That's how I justify next year's budget."
— Jordan Chen, Senior Events Manager

Frustration

No way to track photo engagement or ROI. Privacy concerns with employee photos. Can't prove event value with hard data. Post-event follow-up has low response rates. Budget justification difficult without metrics Leadership asks: "What did we get for that investment?"

How FaceFind Helps

Private events with email whitelist Detailed analytics dashboard Exportable reports for leadership Automatic session expiration Corporate branding integration GDPR/CCPA compliant 73% engagement in 48 hours

Demographics

Name : **Jordan**

Age : 40

Location : Chicago, IL

Company : Global Tech Solutions

Position : Senior Events Manager

Income : \$82,000/year

MVP

MINIMUM VIABLE PRODUCT

Build. Launch. Learn.

● FaceFind · v1.0



CORE FEATURE
AI face recognition
to find your photos



FAST DELIVERY
Ship, iterate, scale
from day one



VALIDATED
Real users, real data,
real decisions



IDEA

MVP

GROWTH

SCALE

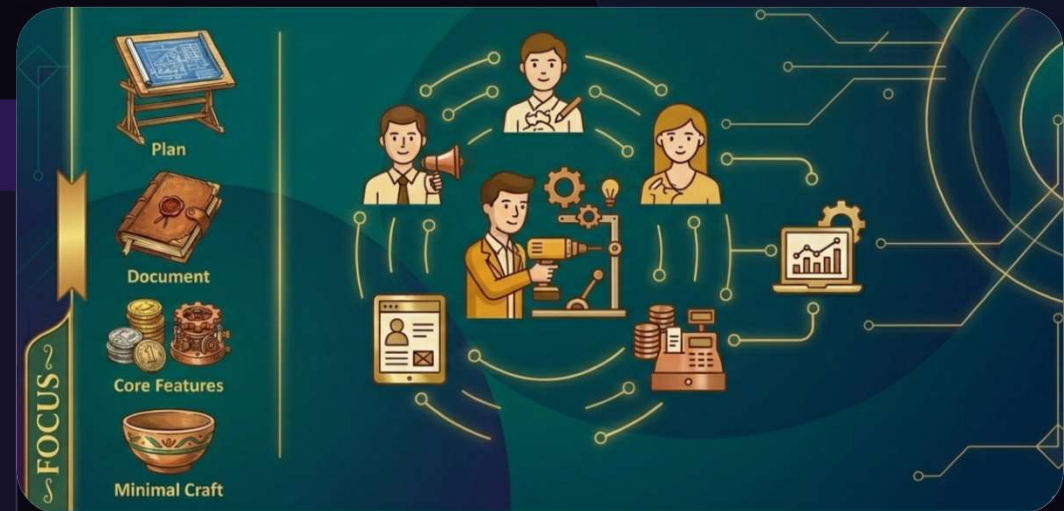
Minimum Viable Product.

What's in..



IN THE MVP

- Photographer register & login (JWT auth)
- Create & manage events with QR codes
- Upload photos → AI face indexing
- Guest selfie → instant photo match
- Real-time processing dashboard (WebSocket)
- Event catalog & browsing



Full-Stack Technology Choices.

Frontend

- React.js
- Redux Toolkit
- Tailwind CSS

Backend

- Node.js + Express
- MongoDB + Redis
- JWT Auth
- Socket.io

Machine Learning

- Python + PyTorch
- FastAPI
- InsightFace
- OpenCV + NumPy

Cloud & DevOps

- AWS S3 + CloudFront CDN
- Docker Containers
- GitHub Actions CI/CD
- Nginx Reverse Proxy

ALGORITHMS

ML / AI Algorithms Used.



ML / AI Algorithms

Core framework



Face Detection

MTCNN / RetinaFace



Face Embedding

ArcFace (InsightFace)



Similarity Search

FAISS / pgvector



Clustering (Future)

K-Means / DBSCAN

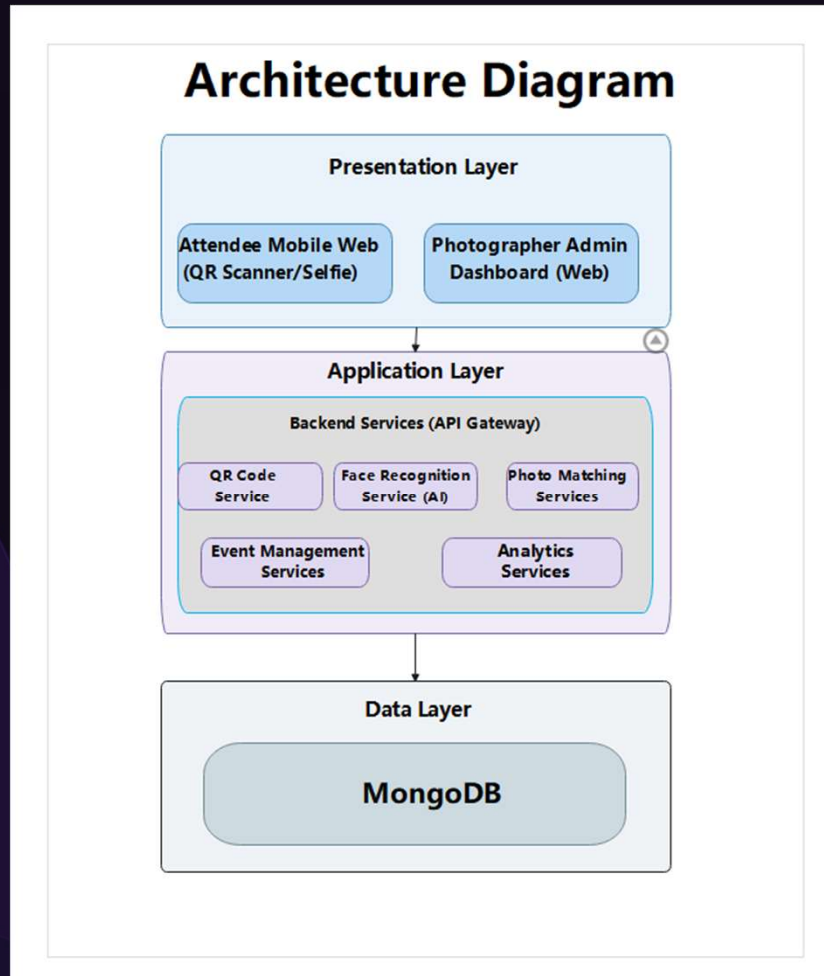


Pattern Recognition

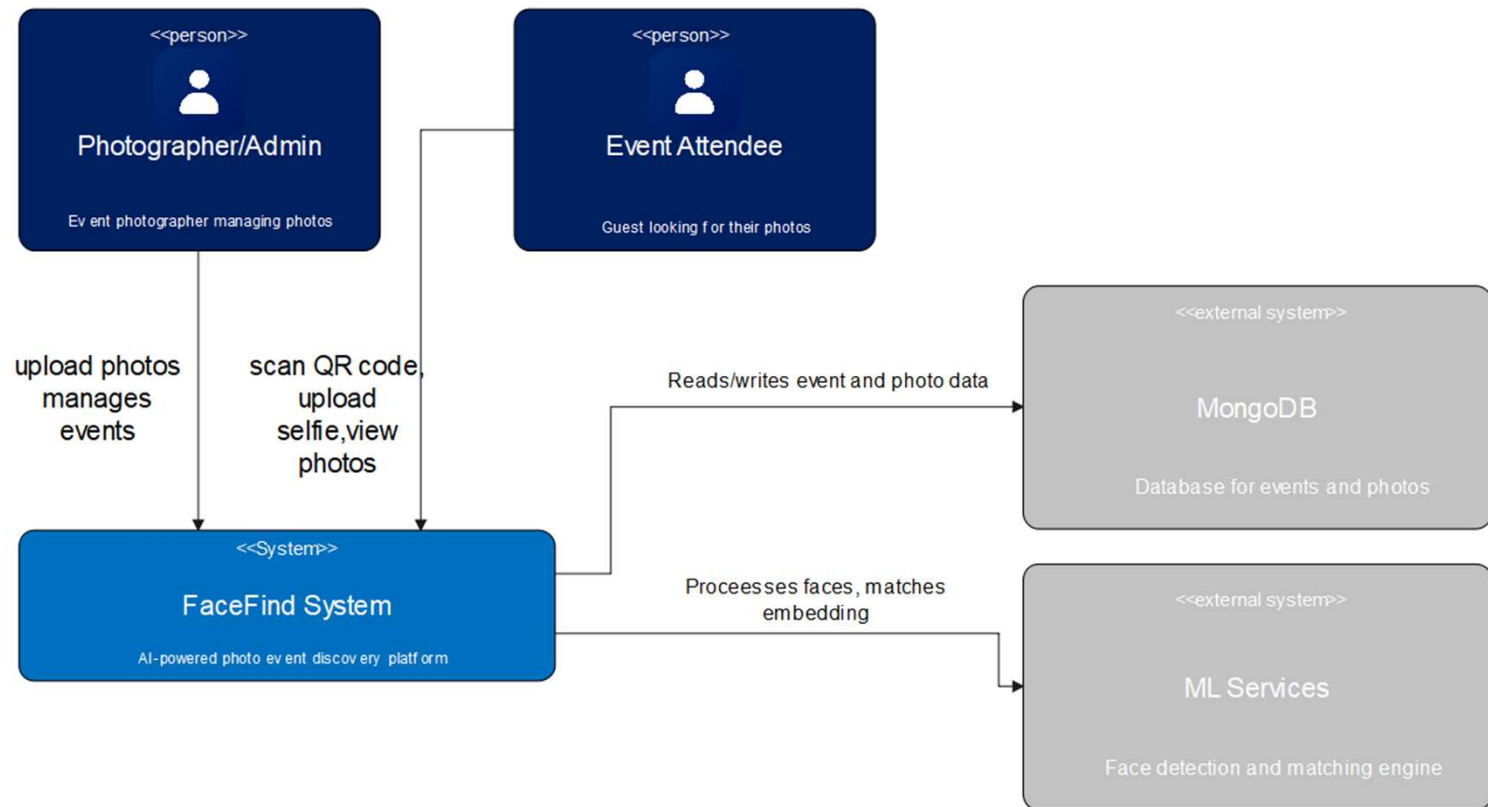
Deep Learning CNNs

DIAGRAMS

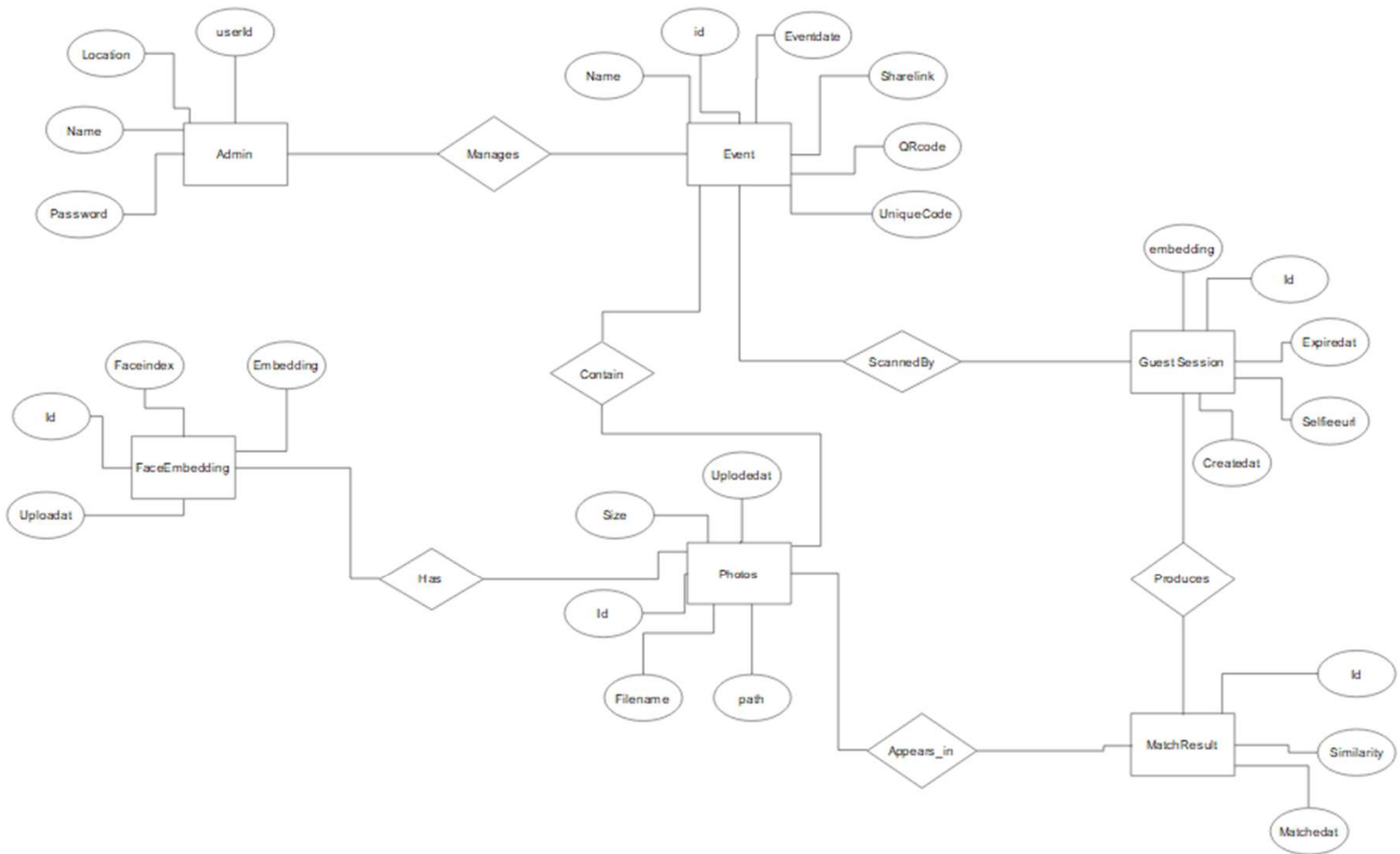
Architecture Diagram



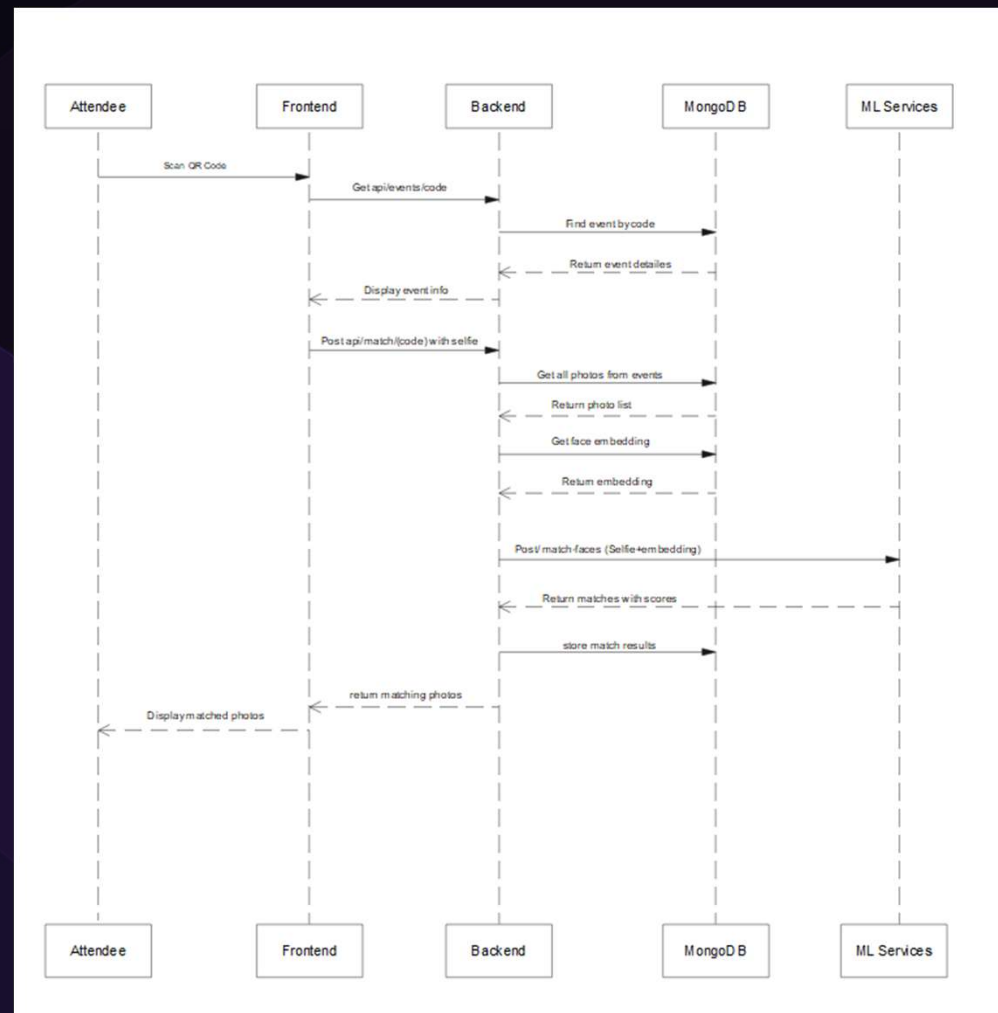
Context Diagram



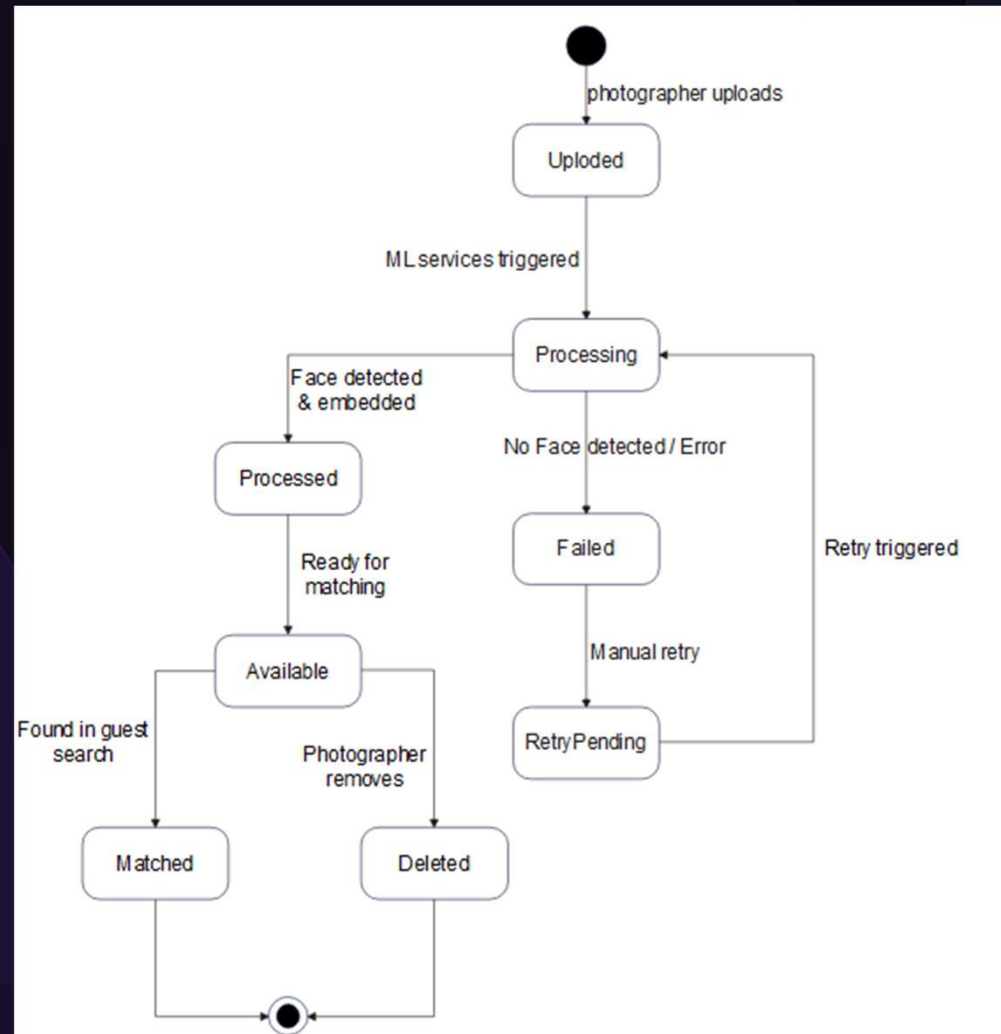
ER Diagram



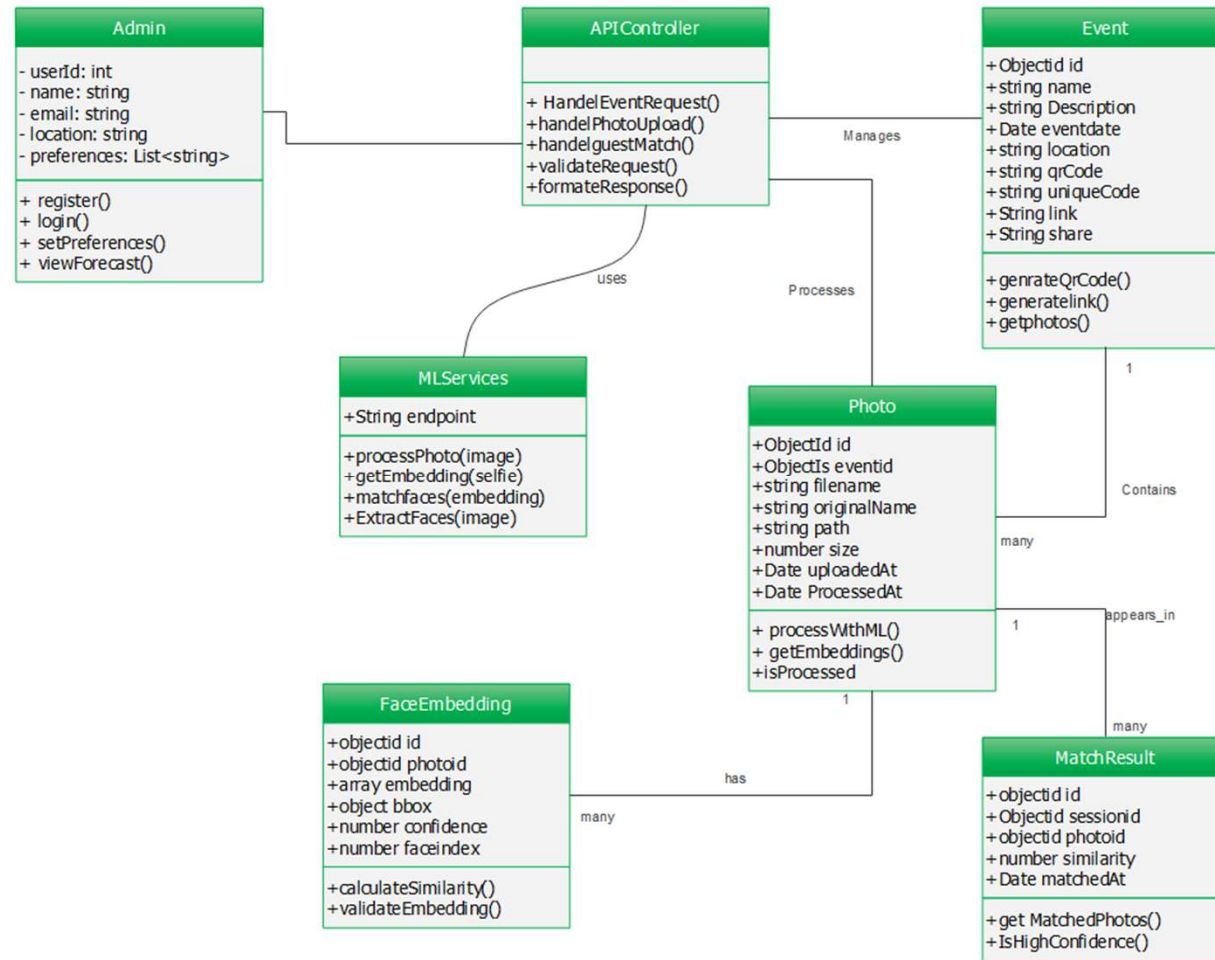
Sequence Diagram



State Diagram



Class Diagram



PRODUCT BACKLOG

PRIORITIZED LIST OF DELIVERABLES

Sprint 1 Backlog

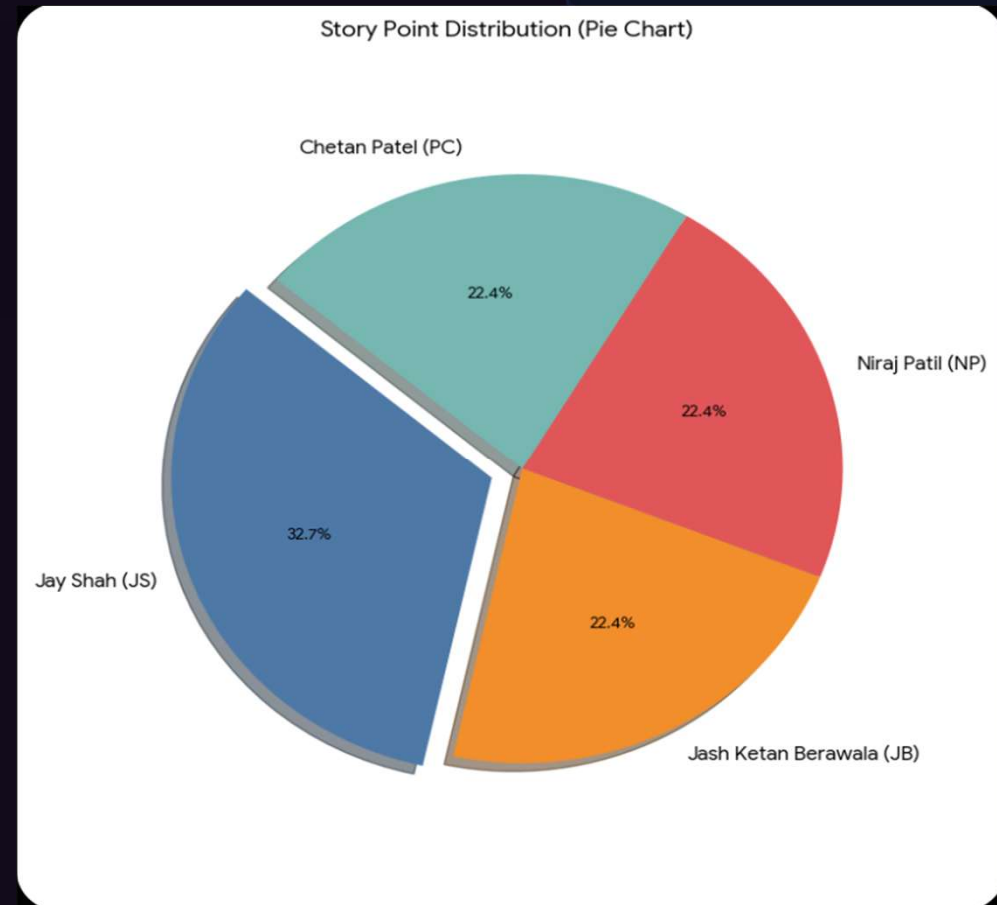
☐ Capstone Sprint 2 Add dates (10 work items)	0	0	49	Start sprint	..
☒ FF-17 Sprint 2 backlock investigatio...	DONE ▾	3		NP	
☒ FF-18 Part 1: Preparing presentable...	DONE ▾	3		JB	
☒ FF-20 jira work	DONE ▾	8		PC	
☒ FF-21 Preparing Personas	DONE ▾	5		JB	
☒ FF-22 Preparing Charts	DONE ▾	5		NP	
☐ FF-24 Designing the frontend and E...	DONE ▾	8	=	JS	
☐ FF-25 Environment Initialization and...	DONE ▾	8	=	JS	
☐ FF-28 Part 2: Preparing presentable...	DONE ▾	3	=	JB	
☐ FF-27 Investigation for next sprint d...	DONE ▾	3	=	PC	
☐ FF-26 Metrics	DONE ▾	3	=	NP	

Sprint Velocity Metrics

A team's velocity chart tracks the amount of work completed in each sprint over a series of sprints, enabling the identification of trends and the prediction of future capacity. Velocity typically reflects the story points or work units a team can realistically commit to and successfully complete in a given iteration, a concept further explained by .

Here is the velocity data for "Capstone Sprint 2":

- **Sprint Name:** Capstone Sprint 2
- **Total Committed Points:** 49
- **Total Completed Points:** 49
- Zero Spillover
- **Velocity:** 49
- Team Member Velocity Chart



Sprint burndown



49 points done, 0 points to go



Technical Cards

 Add epic /

 FF-24

**Frontend Development –
Landing Page and Event
Catalog**

Done ▾

✓ Done



 Improve Story

▼ **Key details**

Priority

 Medium

 Add epic /

 FF-25

**Implement Login
Authentication API using
Node.js**

Done ▾

✓ Done



 Improve Story

▼ **Key details**

Priority

 Medium

SPRINT BACKLOG

A prioritised list of work for the current Sprint



IMR 9
DAY 10
DAY 5
DAY 1

Sprint 2 Backlog

☐ ▼ **Capstone Sprint 3**  Add dates (7 work items) 56 0 0 Start sprint ...

☐  FF-30 ML Service Connectivity	TO DO ▼	8		
 FF-32 Real-time Processing Feedback	TO DO ▼	8		
 FF-34 Face Detection Pipeline	TO DO ▼	8		
 FF-31 Embedding Extraction	TO DO ▼	8		
 FF-33 Guest Landing Page	TO DO ▼	8		
 FF-36 Unique Event Logic	TO DO ▼	8		
 FF-35 Basic Image Upload	TO DO ▼	8		

+ Create

Sprint 3 Backlog

☐ ▼ **Capstone Sprint 4**  Add dates (7 work items)

45 0 0 Start sprint ...

 FF-29	Selfie Processing Logic	TO DO ▼	8	=	
 FF-37	Similarity Matching Engine	TO DO ▼	8	=	
 FF-38	Personalized Results Gallery	TO DO ▼	8	=	
 FF-39	Download & Sharing	TO DO ▼	8	=	
 FF-40	Full-Stack Deployment	TO DO ▼	8	=	
☐ ☒ FF-41	Sprint 3 Deliverables 	+ Epic TO DO ▼	5		 ...
☒ FF-42	CLONE - Sprint 4 Deliverables	TO DO ▼	-		

+ Create

Test Cases

FaceFind_Test_Cases

Test case id	Unit to test	Assumptions	Test data	Steps to be executed	Expected Results	Actual Result	Pass/Fail	Comments
FF-UI-01	Frontend: Photographer Login & Dashboard Access	User is registered; Backend server is running.	Email: user@test.com, Password: Password123	1. Navigate to /login 2. Enter valid credentials 3. Click 'Login'	User is redirected to the Dashboard page; Photographer-specific events are displayed.	Successfully redirected to dashboard; Photographer view loaded without errors.	Pass	Core frontend entry point
FF-BE-01	Backend: Event Initialization	Database connection is active.	JSON Body: { 'eventName': 'Summer Gala 2024', 'eventDate': '2024-07-20' }	1. Send POST request to /api/events	HTTP 201 Created; Event metadata is successfully stored in MongoDB.	Event 'Summer Gala 2024' record found in MongoDB; API returned success.	Pass	Core backend event storage

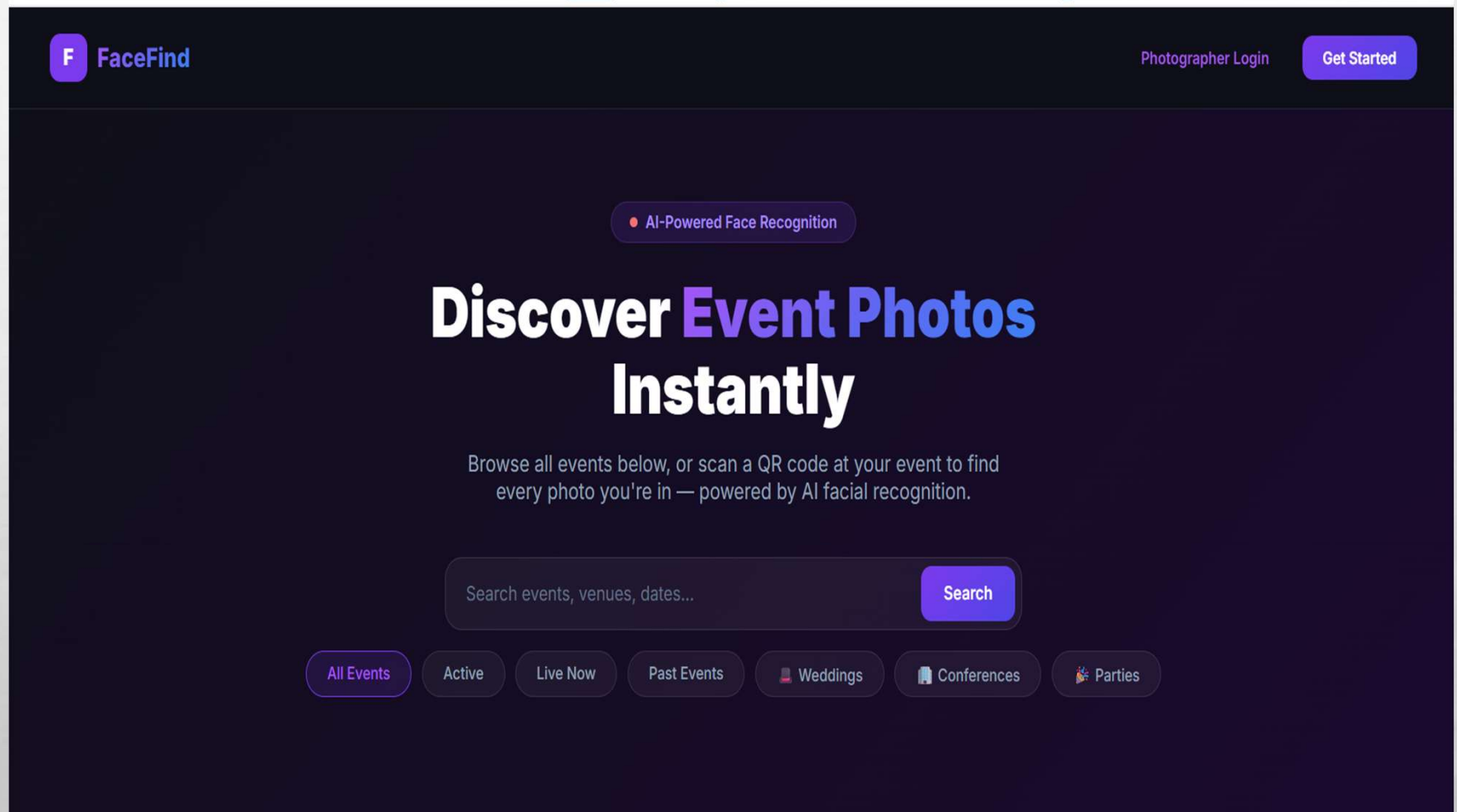


Sprint 1

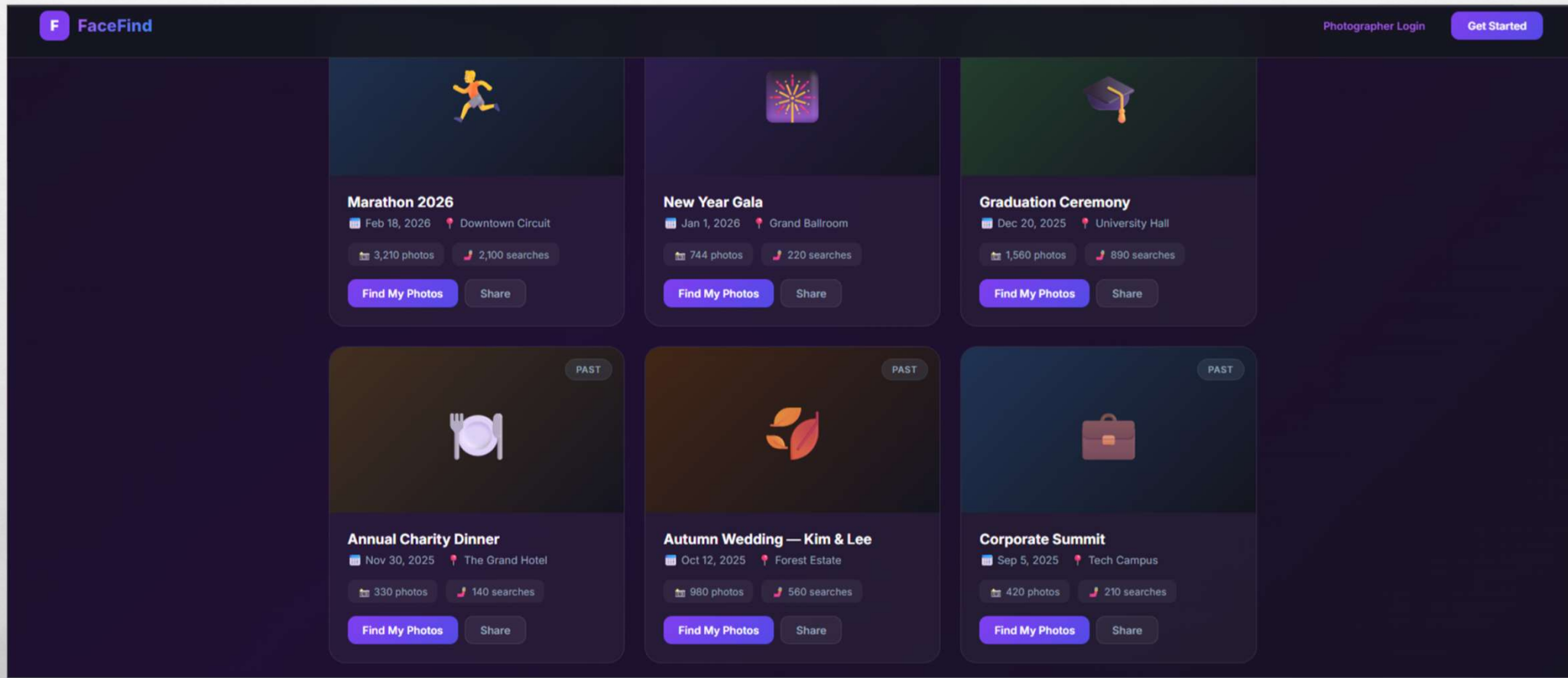
Project Demo



First Page (Demo)



Catalog Page



Register Page

Sign In

Create Account

FIRST NAME

Jash

LAST NAME

B

EMAIL ADDRESS

jk@gmail.com

STUDIO / BUSINESS NAME

Jk Studios

PASSWORD

.....



CONFIRM PASSWORD

.....

☐ I agree to the [Terms of Service](#) and [Privacy Policy](#)

Create My Account

Already have an account? [Sign in](#) →

Login page

F

FaceFind

Photographer Login

Get Started

← Back to Catalog

F

FaceFind

Photographer Portal

Sign in or create your photographer account

Sign In

Create Account

EMAIL ADDRESS

you@studio.com

PASSWORD

☐ Remember me

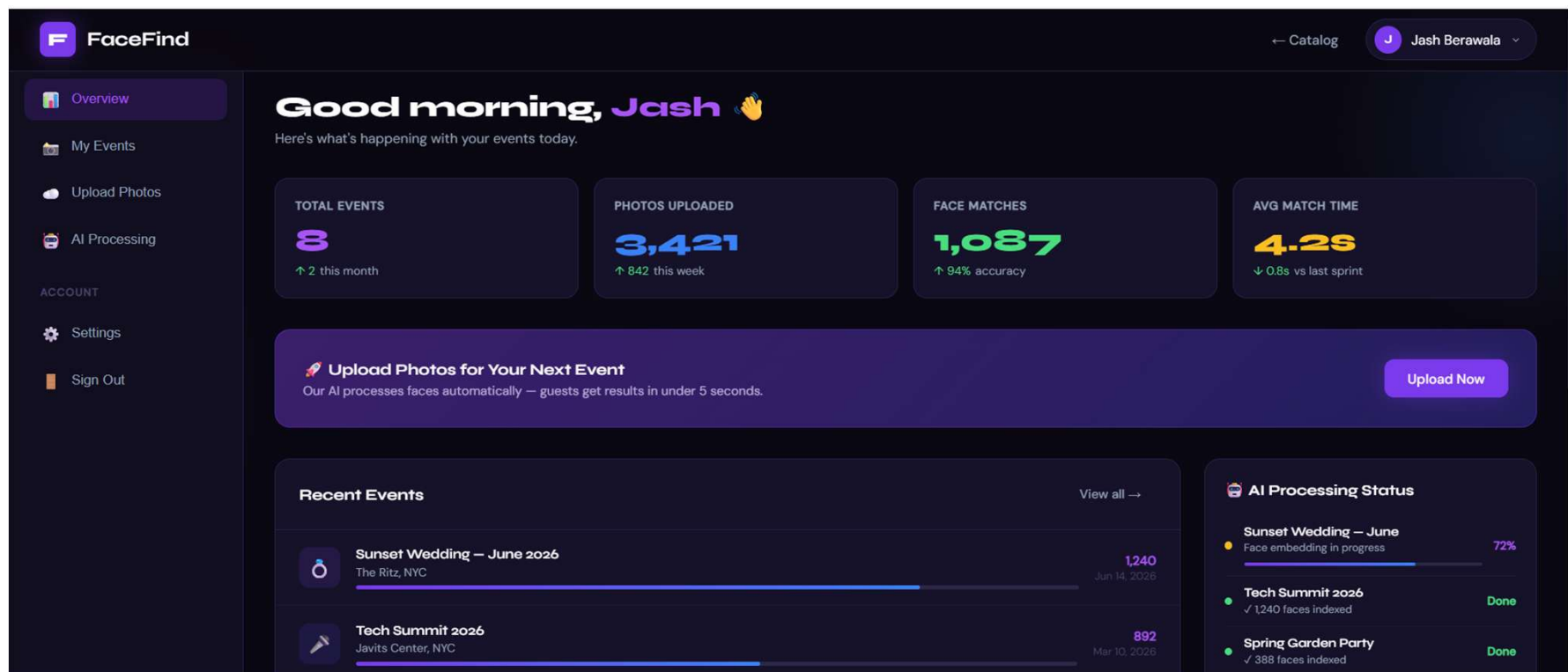
[Forgot password?](#)

Sign In to Dashboard

No account yet? [Create one free](#) →

Powered by GEE, a 3D environment. Your data stays on-site.

Dashboard Page (after Login)



Sprint 1 Retrospective

WENT WELL

- 0 spillover & 100% tasks achieved
- Optimized 3-day deep-dive sessions
- Strong team synergy & respect
- Improved Jira review workflow

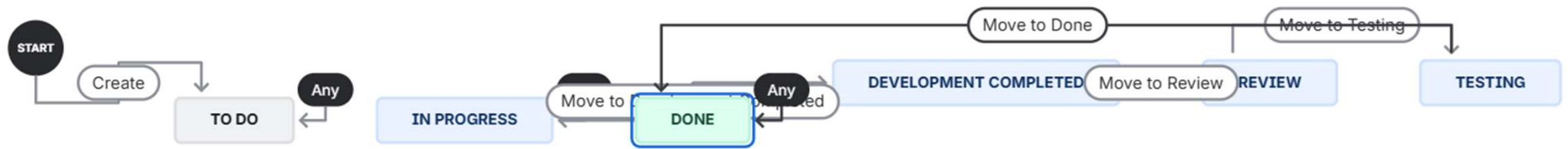
IMPROVE

- Backlog grooming sessions
- Refinement of User Personas

ACTION ITEMS

- Schedule dedicated grooming
- Update persona documents
- Formalize 3-day session schedule

New Jira Workflow





Thank You!

Questions welcome.



Group Wiki

github.com/JashBerawala/Pace-Project-Team7/wiki



GitHub Repo

github.com/JashBerawala/Pace-Project-Team7



Team 7

CS 691 · Pace University